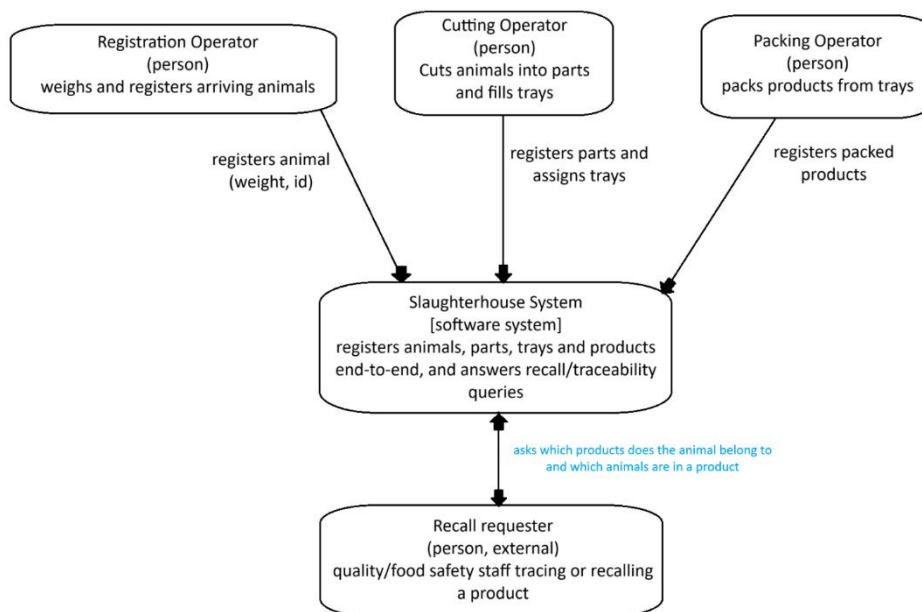


PRO 3 – Assignment 2

1. C1 – System context view

The system context view is mostly the same as in part 1. In part 2, the requester's questions become more concrete.

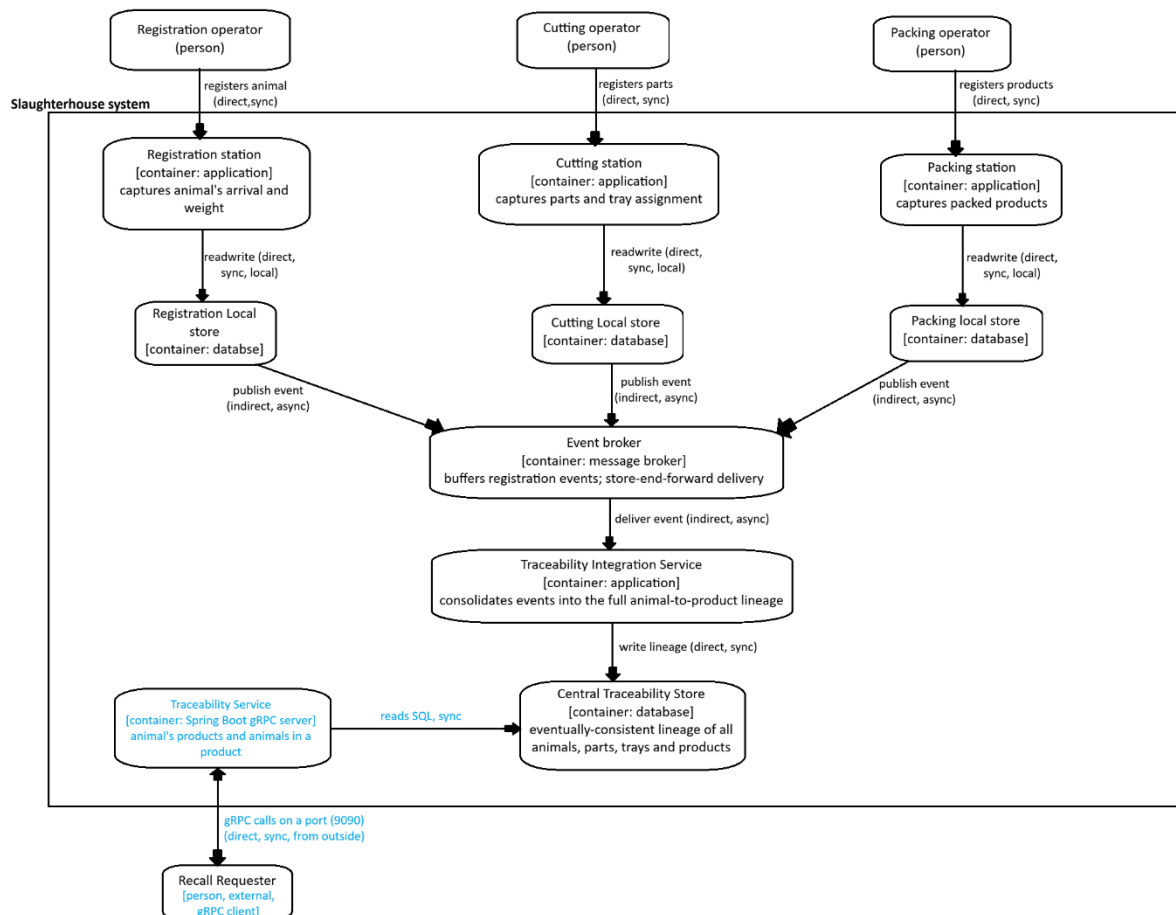


(the blue text shows what has changed)

Actor	Type	Role
Registration Operator	Person, internal	Weighs and registers each arriving animal.
Cutting Operator	Person, internal	Cuts animals into parts and assigns parts to trays.
Packing Operator	Person, internal	Packs products (part packages or “half animals”) from tray contents.
Recall Requester	Person, external	Quality/food-safety staff who need to trace an animal to the products it could be part of, from outside the slaughterhouse.

2. C2 – Container View

The container view is mostly the same as in part 1. The Recall Service is now a Traceability Service, Spring Boot gRPC server.



Container	Kind	Responsibility
Registration / Cutting / Packing Station App	Application	The program an operator uses at that station. Writes only to its own local store, then goes on with its work.
Registration / Cutting / Packing Local Store	Database	Local source of truth for one station. Always writable regardless of network state.
Event Broker	Message Broker	Buffers “{Animal Part Product} registered” events published by the

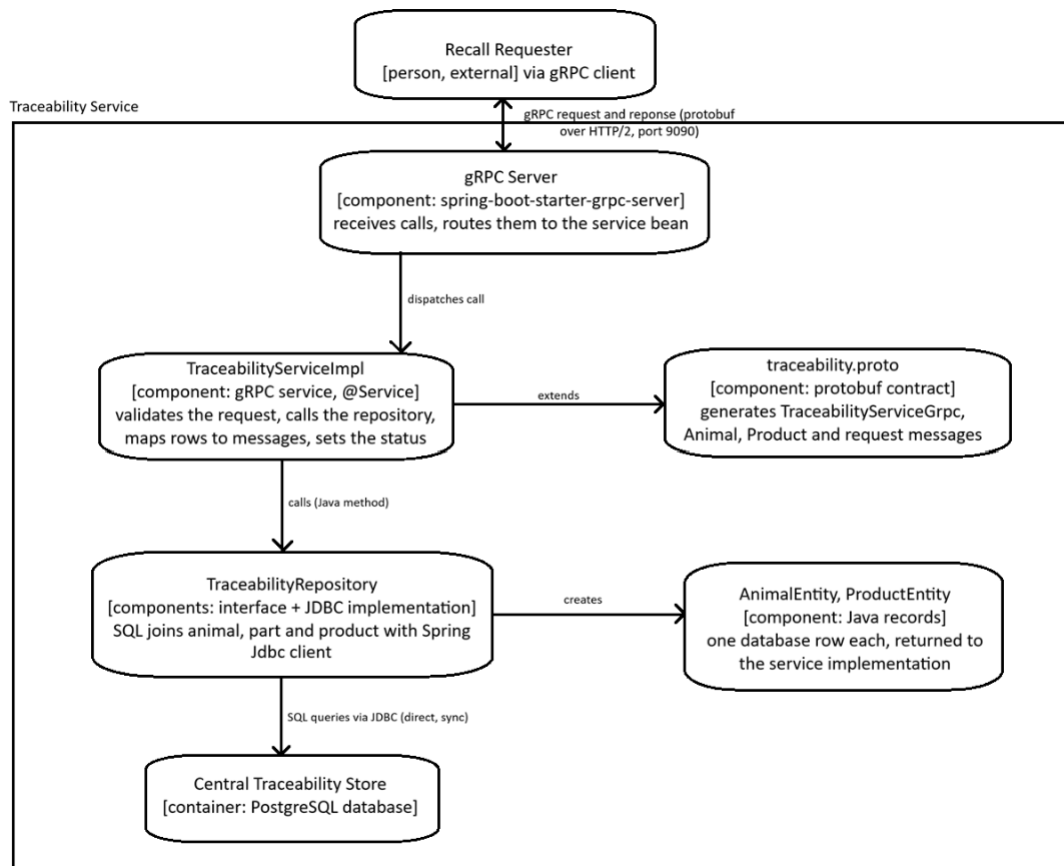
		stations and forwards them when the receiving side is reachable.
Traceability Integration Service	Application	Consumes events into one consistent, cross-station lineage.
Central Traceability Store	Database	Eventually-consistent record of the full Animal → Part → Tray → Product lineage.
Traceability Service	Spring Boot gRPC server	Answers 'animals in a product' and 'products of an animal' from outside the slaughterhouse.

Types of communication:

Interaction	Type	Timing	Technology
Operator → Station App	Direct	Synchronous	Local UI
Station App → Local Store	Direct	Synchronous	Local, no network
Station App → Event Broker	Indirect	Asynchronous	Open
Event Broker → Integration Service	Indirect	Asynchronous	Open
Integration Service → Central Store	Direct	Synchronous	SQL
Traceability Service → Central Store	Direct	Synchronous	SQL over JDBC
Recall Requester ↔ Traceability Service	Direct	Synchronous	gRPC (remote procedure call, protobuf over HTTP/2)

3. C3 – Component View

The component view focuses on Traceability Service. A call goes from gRPC server to the service implementation, then to repository, which runs SQL.



Components:

Component	Technology	Responsibility
gRPC Server	spring-boot-starter-grpc-server	Listens on port 9090 and routes each call to the registered service bean.
Traceability.proto	Protocol Buffers, protobuf - maven	The contract. Generates traceabilityServiceGrpc and the message classes into package 'generated'
TraceabilityServiceImpl	Java, extends TraceabilityServiceImplBase	Checks the request, asks the repository, maps entities to proto messages and returns errors when needed.

TraceabilityRepository	Java interface + TraceabilityRepositoryJdbc	Runs the SQL queries. Hides the database from gRPC layer
AnimalEntity, ProductEntity	Java records	Each object holds data of one row from database